CHETHAN KUMAR B S

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PROFESSIONAL SUMMARY

Full Stack Engineer and Computer Science undergraduate (2027 Batch) with hands-on expertise in engineering high-throughput distributed web applications, event-driven microservices, and secure backend architectures using React.js, Next.js, Node.js, and PostgreSQL. Experienced in implementing cryptographic security protocols (AES-256), optimizing complex relational databases (ACID transactions, B-tree indexing, Redis caching), and designing responsive, high-concurrency systems. Grounded in core Computer Science fundamentals (DSA, DBMS, OS, Computer Networks) with a passion for building scalable, production-ready full-stack software.

TECHNICAL SKILLS

- **Languages:** JavaScript (ES6+), TypeScript, Python, SQL, C++, C, HTML5, CSS3
- **Frontend Development:** React.js, Next.js 14, Redux Toolkit, Tailwind CSS, Responsive Web Design, SSR/SSG, WebSockets
- **Backend & APIs:** Node.js, Express.js, RESTful API Design, Socket.io, Microservices Architecture, JWT, RBAC, Middleware
- **Databases & Caching:** PostgreSQL, MySQL, Redis, Database Normalization (3NF), Query Optimization, Indexing, Connection Pooling
- **DevOps & Tools:** Git, GitHub, Docker, Postman, Linux/Bash, Vite, VS Code, CI/CD Basics, Vercel
- **Core CS & Security:** Data Structures & Algorithms, Object-Oriented Programming (OOP), DBMS, Operating Systems, Computer Networks, AES-256 Encryption, OWASP Web Security

KEY ENGINEERING PROJECTS

PRISM Platform for Real-time Intelligence & Secure Management **Next.js, Node.js, PostgreSQL, Redis, WebSockets, Docker** *Final Year Capstone Project Full Stack Enterprise Monitoring & Security Platform* *GitHub: StratagemVX/PRISM*

- Architected an end-to-end full-stack intelligence platform enabling real-time telemetry streaming, role-based access control (RBAC), and automated threat detection workflows.
- Engineered high-throughput Node.js microservices integrated with Socket.io/WebSockets, achieving sub-45ms bidirectional state synchronization across 500+ simulated concurrent active sessions.
- Designed normalized PostgreSQL relational schemas coupled with a Redis in-memory caching tier, slashing repetitive analytical query response latency by 65%.
- Implemented AES-256-GCM encrypted payload channels and JWT session verification with rate-limiting middleware, ensuring strict OWASP API compliance.
- Containerized microservices using Docker and orchestrated unified developer environments with structured logging and environment isolation.

Self-Destructive Encrypted Messaging System **React.js, Node.js, Express.js, PostgreSQL, Web Crypto API**
Production-Ready Ephemeral Communication Architecture *GitHub: StratagemVX/secure-messenger*

- Built a zero-knowledge, privacy-focused messaging application executing client-side end-to-end encryption via AES-256-GCM algorithms.
- Designed automated Time-To-Live (TTL) expiration protocols and asynchronous purge cron routines to irreversibly wipe expired message data and metadata from PostgreSQL storage.
- Developed an interactive React interface featuring real-time self-destruct timers, dynamic read receipts, and defensive sanitization against Cross-Site Scripting (XSS).
- Configured backend RESTful endpoints with CORS policies, Helmet security headers, and connection pooling (`pg-pool`) sustaining high request throughput.

High-Concurrency Railway Reservation & Transit Engine **React.js, Node.js, Express.js, PostgreSQL, SQL**
Enterprise Database Management System & Administrative Portal *GitHub: StratagemVX/railway-dbms*

- Designed an enterprise relational database architecture modeling train schedules, dynamic routes, passenger manifests, and tiered seat inventory.
- Engineered ACID-compliant SQL transactions and row-level locking mechanisms, eliminating race conditions and over-booking anomalies during simultaneous seat reservations.
- Optimized multi-table JOIN operations across 100,000+ synthetic passenger records using composite B-tree indexes, reducing overall query latency from 320ms to 42ms.
- Constructed an administrative management dashboard with React and REST APIs for real-time dispatch monitoring and automated booking analytics.

AI-Powered Developer Intelligence Portfolio **Next.js 14, React, TypeScript, Tailwind CSS, REST API** *Modern Responsive Web Platform with Dynamic Integrations* *GitHub: StratagemVX/portfolio*

- Engineered an AI-themed interactive developer portfolio utilizing Next.js 14 Server-Side Rendering (SSR) and Incremental Static Regeneration (ISR), achieving a 99+ Google Lighthouse performance rating.
- Integrated GitHub REST API to dynamically fetch and display live repository metrics, commit velocity, and deployment statistics.
- Implemented modular Tailwind CSS component architectures with fluid dark-mode transitions, responsive layouts, and zero Cumulative Layout Shift (CLS).

EDUCATION

Bachelor of Engineering in Computer Science and Engineering *Expected 2027*
MVJ College of Engineering, Bengaluru *Bengaluru, India*
Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems (DBMS), Operating Systems, Computer Networks, Software Engineering, Web Technologies.

TECHNICAL ACHIEVEMENTS & CERTIFICATIONS

- **Algorithmic Problem Solving:** Solved 200+ Data Structures & Algorithms problems across LeetCode and GeeksforGeeks, focusing on Graph Algorithms, Dynamic Programming, Trees, and Array manipulation.
- **Engineering Leadership:** Spearheaded full-stack architecture, database modeling, and cryptographic integration for final year capstone project **PRISM**.
- **Certifications:** Full Stack Web Development (React, Node.js), Python for Software Systems, SQL & Relational Database Design, Data Structures & Algorithms in C++.